

NUMBER SERIES

MULTIPLICATION TABLES (2-50)(MUST LEARN)

2	2, 4, 6, 8, 10, 12, 14, 16, 18, 20
3	3, 6, 9, 12, 15, 18, 21, 24, 27, 30
4	4, 8, 12, 16, 20, 24, 28, 32, 36, 40
5	5, 10, 15, 20, 25, 30, 35, 40, 45, 50
6	6, 12, 18, 24, 30, 36, 42, 48, 54, 60
7	7, 14, 21, 28, 35, 42, 49, 56, 63, 70
8	8, 16, 24, 32, 40, 48, 56, 64, 72, 80
9	9, 18, 27, 36, 45, 54, 63, 72, 81, 90
10	10, 20, 30, 40, 50, 60, 70, 80, 90, 100
11	11, 22, 33, 44, 55, 66, 77, 88, 99, 110
12	12, 24, 36, 48, 60, 72, 84, 96, 108, 120
13	13, 26, 39, 52, 65, 78, 91, 104, 117, 130
14	14, 28, 42, 56, 70, 84, 98, 112, 126, 140
15	15, 30, 45, 60, 75, 90, 105, 120, 135, 150
16	16, 32, 48, 64, 80, 96, 112, 128, 144, 160
17	17, 34, 51, 68, 85, 102, 119, 136, 153, 170
18	18, 36, 54, 72, 90, 108, 126, 144, 162, 180
19	19, 38, 57, 76, 95, 114, 133, 152, 171, 190
20	20, 40, 60, 80, 100, 120, 140, 160, 180, 200
21	21, 42, 63, 84, 105, 126, 147, 168, 189, 210
22	22, 44, 66, 88, 110, 132, 154, 176, 198, 220
23	23, 46, 69, 92, 115, 138, 161, 184, 207, 230
24	24, 48, 72, 96, 120, 144, 168, 192, 216, 240
25	25, 50, 75, 100, 125, 150, 175, 200, 225, 250
26	26, 52, 78, 104, 130, 156, 182, 208, 234, 260
27	27, 54, 81, 108, 135, 162, 189, 216, 243, 270
28	28, 56, 84, 112, 140, 168, 196, 224, 252, 280
29	29, 58, 87, 116, 145, 174, 203, 232, 261, 290
30	30, 60, 90, 120, 150, 180, 210, 240, 270, 300
31	31, 62, 93, 124, 155, 186, 217, 248, 279, 310
32	32, 64, 96, 128, 160, 192, 224, 256, 288, 320
33	33, 66, 99, 132, 165, 198, 231, 264, 297, 330
34	34, 68, 102, 136, 170, 204, 238, 272, 306, 340
35	35, 70, 105, 140, 175, 210, 245, 280, 315, 350

36	36, 72, 108, 144, 180, 216, 252, 288, 324, 360
37	37, 74, 111, 148, 185, 222, 259, 296, 333, 370
38	38, 76, 114, 152, 190, 228, 266, 304, 342, 380
39	39, 78, 117, 156, 195, 234, 273, 312, 351, 390
40	40, 80, 120, 160, 200, 240, 280, 320, 360, 400
41	41, 82, 123, 164, 205, 246, 287, 328, 369, 410
42	42, 84, 126, 168, 210, 252, 294, 336, 378, 420
43	43, 86, 129, 172, 215, 258, 301, 344, 387, 430
44	44, 88, 132, 176, 220, 264, 308, 352, 396, 440
45	45, 90, 135, 180, 225, 270, 315, 360, 405, 450
46	46, 92, 138, 184, 230, 276, 322, 368, 414, 460
47	47, 94, 141, 188, 235, 282, 329, 376, 423, 470
48	48, 96, 144, 192, 240, 288, 336, 384, 432, 480
49	49, 98, 147, 196, 245, 294, 343, 392, 441, 490
50	50, 100, 150, 200, 250, 300, 350, 400, 450, 500
Number	Table (×1 to ×10)

What is a Series?

A **Series** is a sequence of numbers arranged according to a fixed rule or pattern.

👉 Your task in the exam is to identify that rule and find the missing or next number.

Example

2, 4, 6, 8, ?

Pattern → +2

Answer → 10

What is Number Series? Learn from someone who has already achieved what you are preparing for.

A **Number Series** is a sequence of numbers that follows a logical mathematical rule.

The rule may involve

- Addition (+)
- Subtraction (−)
- Multiplication (×)
- Division (÷)
- Squares
- Cubes
- Prime Numbers
- Alternate Pattern
- Combination of Operations

Golden Rule (Most Important)

Before solving any series,

Step 1

Identify the type of series.

Ask yourself:

Is the difference between numbers small?

OR

Are numbers increasing very fast?

If difference is small

➡ Difference Series

If numbers grow rapidly

➡ Multiplication Series

This single step helps solve most questions quickly.

Types of Number Series

There are mainly **2 Types**

1. Difference Series

In this series,

First find the difference between consecutive numbers.

Example

100, 110, 122, 136, 152

Difference

+10

+12

+14

+16

The difference increases slowly.

Therefore,

✅ Difference Series



2. Multiplication Series

In this series,

Each number is obtained by multiplying the previous number.

Example

10, 20, 40, 80, 160

Difference

10

20

40

80

The difference increases rapidly.

Therefore,

✓ Multiplication Series

DIFFERENCE SERIES

Pattern 1 : Constant Addition

Every number is increased by the same number.

Example

4, 8, 12, 16, 20, ?

Difference

+4

+4

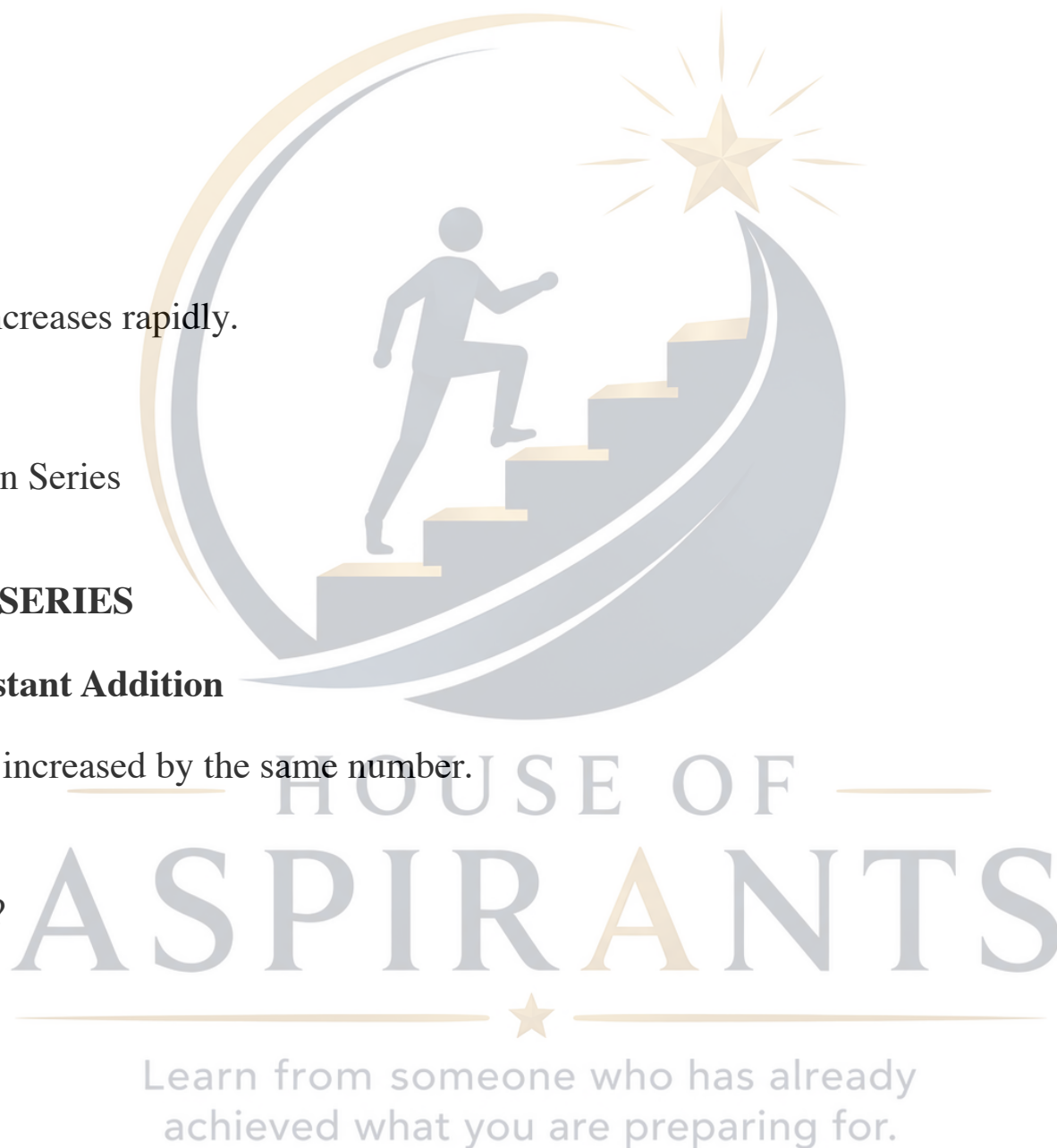
+4

+4

Next

$20 + 4 = 24$

✓ Answer = 24



Pattern 2 : Increasing Addition

The added number keeps increasing.

Example

2, 3, 5, 8, 12, 17, ?

Difference

+1

+2

+3

+4

+5

Next difference = +6

$17 + 6 = 23$

✓ Answer = 23

Pattern 3 : Constant Subtraction

Every number decreases by the same value.

Example

48, 43, 38, 33, 28, ?

Difference

-5

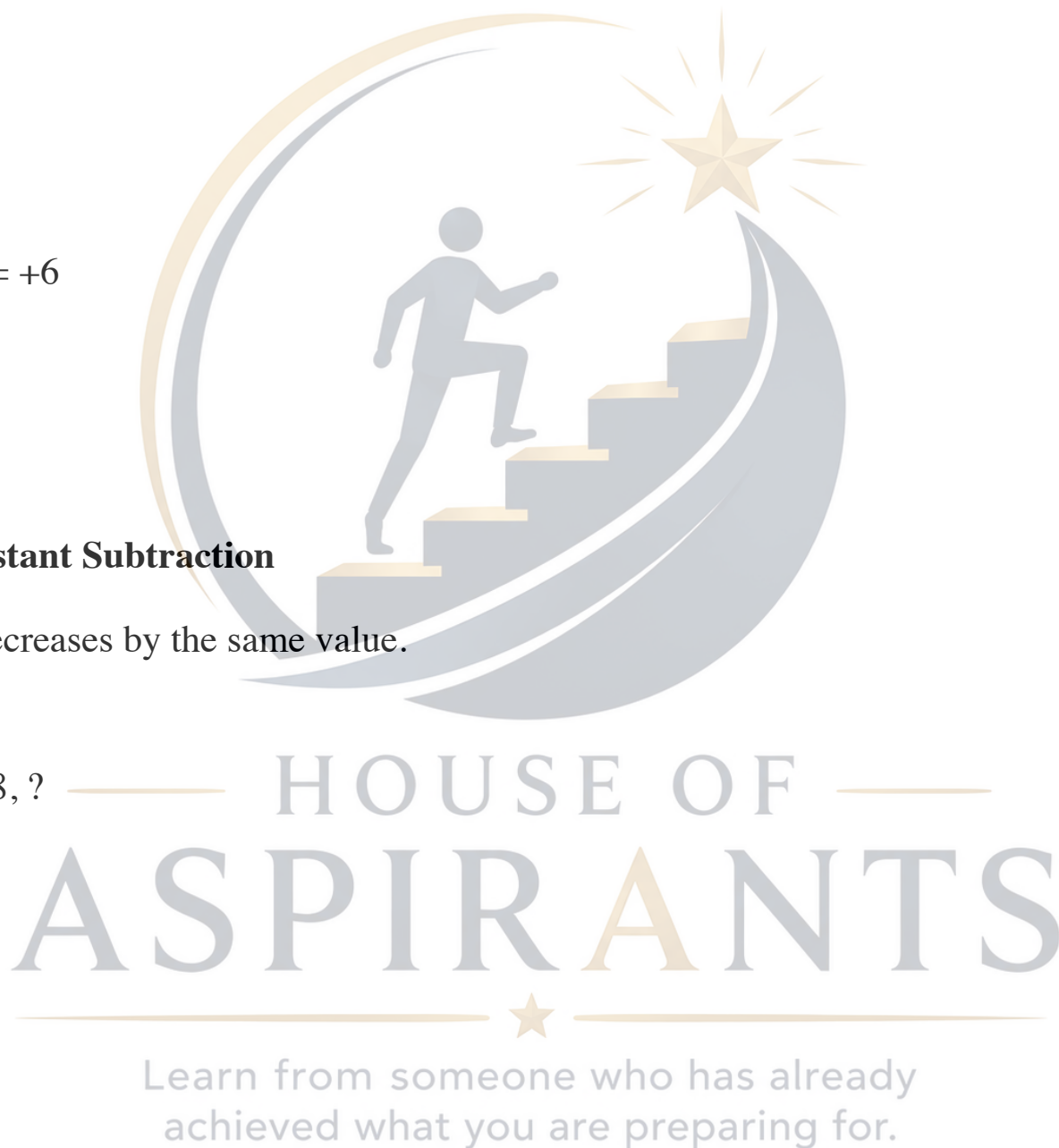
-5

-5

-5

$28 - 5 = 23$

✓ Answer = 23



Pattern 4 : Increasing Subtraction

Example

95, 94, 92, 89, 85, 80, ?

Difference

–1

–2

–3

–4

–5

Next difference

–6

$$80 - 6 = 74$$

✓ Answer = 74

Pattern 5 : Prime Number Series

Prime Number

A number greater than 1 having only two factors

✓ 1 and itself

Prime Numbers (1–100)

Range	Prime Numbers
1–20	2, 3, 5, 7, 11, 13, 17, 19
21–40	23, 29, 31, 37
41–60	41, 43, 47, 53, 59
61–80	61, 67, 71, 73, 79
81–100	83, 89, 97

Example

2, 3, 5, 7, 11, ?

Next Prime

13

✓ Answer = 13

What are Consecutive Numbers?

Consecutive numbers are numbers that come **one after another** in order.

The difference between two consecutive numbers is always **1**.

Formula

Next Number = Previous Number + 1

MULTIPLICATION SERIES

Numbers increase rapidly.

Pattern 1 : Constant Multiplication

Example

4, 12, 36, 108, 324, ?

×3

×3

×3

×3

$324 \times 3 = 972$

✅ Answer = 972

Pattern 2 : Multiplication by Increasing Numbers

Example

3, 3, 4.5, 9, 22.5, ?

Operations

×1

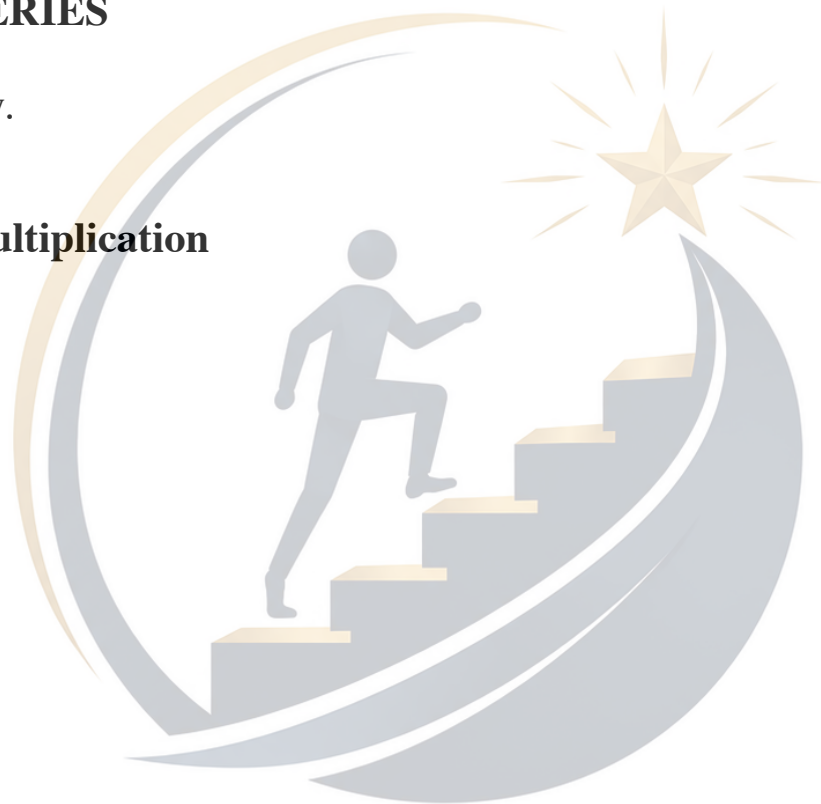
×1.5

×2

×2.5

Next

×3



HOUSE OF
ASPIRANTS

Learn from someone who has already
achieved what you are preparing for.

$$22.5 \times 3 = \mathbf{67.5}$$

✓ Answer = **67.5**

Pattern 3 : Multiply then Add

Formula

Previous Number $\times$ Constant + Fixed Number

Example

5, 11, 23, 47, 95, ?

Rule

$\times 2 + 1$

$95 \times 2 + 1$

=191

✓ Answer = **191**

Pattern 4 : Multiply then Add Consecutive Numbers

Example

5, 11, 24, 51, 106, ?

Rule

$\times 2 + 1$

$\times 2 + 2$

$\times 2 + 3$

$\times 2 + 4$

Next

$106 \times 2 + 5$

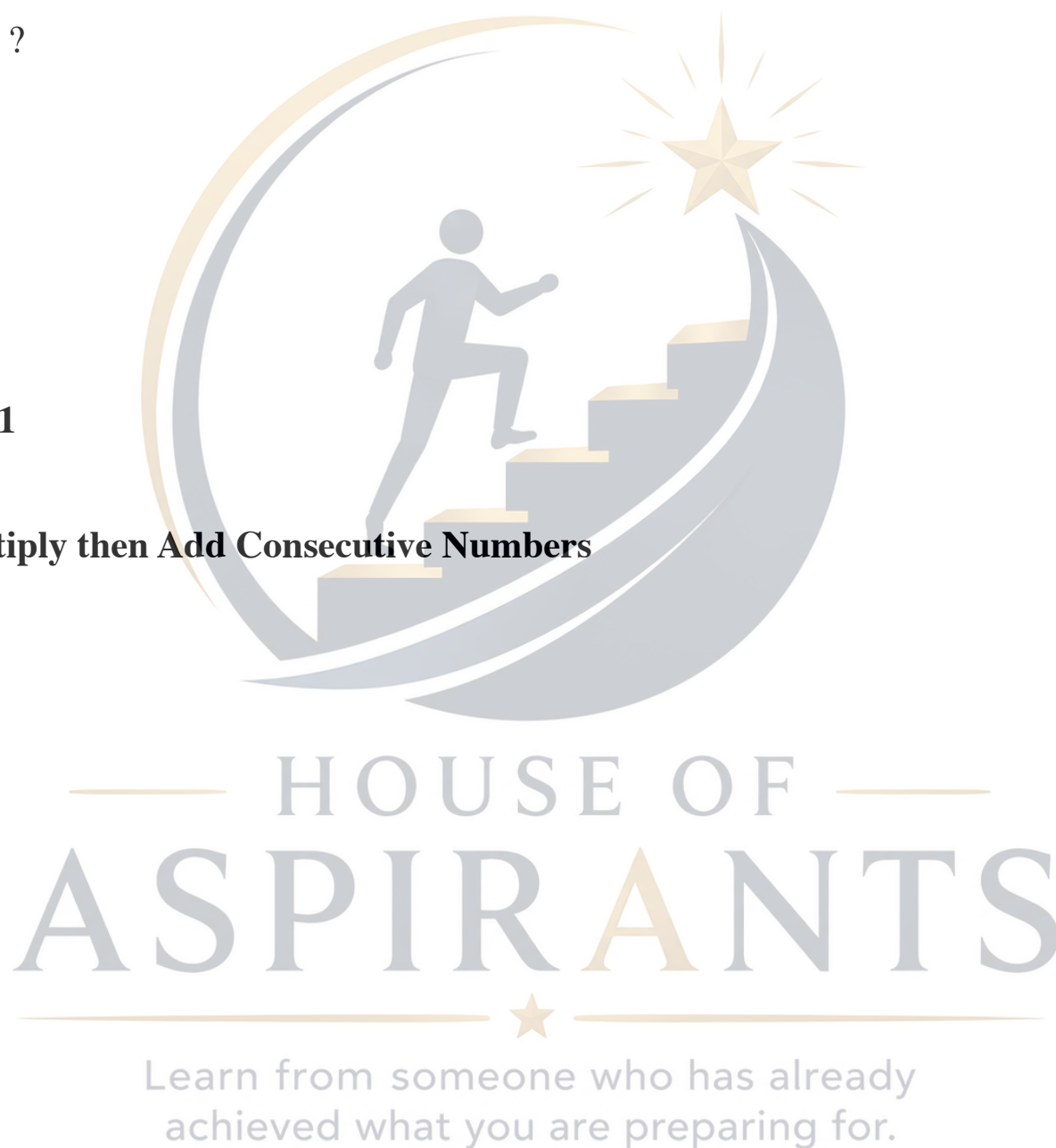
=217

✓ Answer = **217**

Pattern 5 : Increasing Multiplication + Constant Addition

Example

3, 6, 15, 48, 195, ?



Rule

$$\times 1 + 3$$

$$\times 2 + 3$$

$$\times 3 + 3$$

$$\times 4 + 3$$

Next

$$195 \times 5 + 3$$

$$= 978$$

✓ Answer = **978**

Division Series

Numbers decrease by division.

Pattern 1 : Constant Division

Example

$$336, 168, 84, 42, 21, ?$$

$$\div 2$$

$$\div 2$$

$$\div 2$$

$$\div 2$$

$$21 \div 2$$

$$= 10.5$$

✓ Answer = **10.5**



Pattern 2 : Division by Increasing Numbers

Example

$$7200$$

$$3600$$

$$1200$$

$$300$$

60

?

Operations

 $\div 2$ $\div 3$ $\div 4$ $\div 5$

Next

 $\div 6$ $60 \div 6$ $= 10$
 Answer = **10**
Multiplication + Division Series

Numbers increase and decrease alternately.

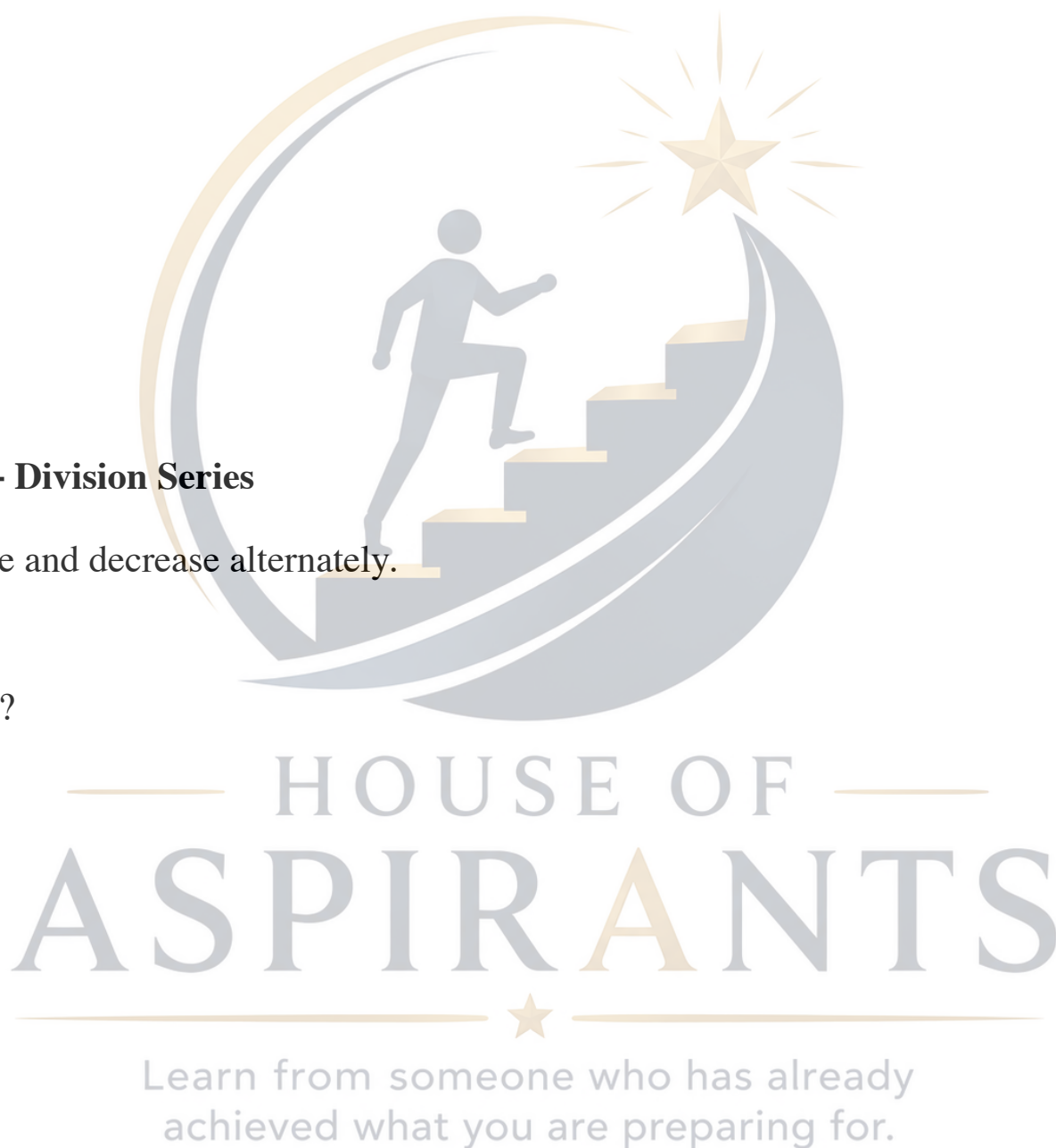
Example

24, 72, 36, 108, 54, ?

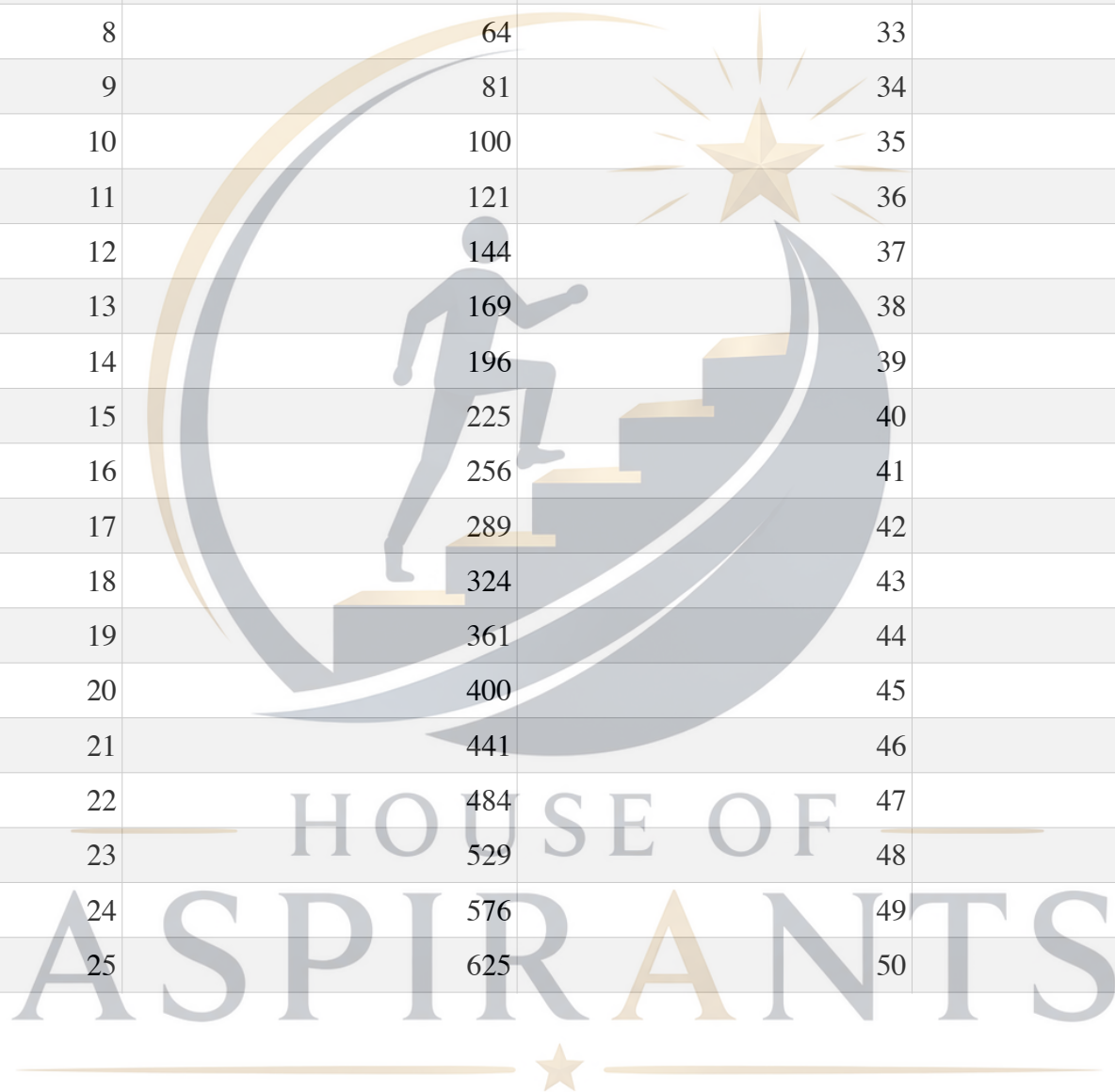
Pattern

 $\times 3$ $\div 2$ $\times 3$ $\div 2$ $\times 3$ 54×3 $= 162$
 Answer = **162**
Square Series

Remember Squares (1–50)



Number	Square	Number	Square
1	1	26	676
2	4	27	729
3	9	28	784
4	16	29	841
5	25	30	900
6	36	31	961
7	49	32	1024
8	64	33	1089
9	81	34	1156
10	100	35	1225
11	121	36	1296
12	144	37	1369
13	169	38	1444
14	196	39	1521
15	225	40	1600
16	256	41	1681
17	289	42	1764
18	324	43	1849
19	361	44	1936
20	400	45	2025
21	441	46	2116
22	484	47	2209
23	529	48	2304
24	576	49	2401
25	625	50	2500



Example

1,4,9,16,25,36,?

Next square

49

✓ Answer = 49

Learn from someone who has already
achieved what you are preparing for.

Difference as Squares

Example

2,3,7,16,32,?

Difference

1

4

9

16

Next difference

25

32+25

=57

✓ Answer = 57



Cube Series

Remember Cubes (1–35)

Number	Cube
1	1
2	8
3	27
4	64
5	125
6	216
7	343
8	512
9	729
10	1000
11	1331
12	1728
13	2197
14	2744
15	3375
16	4096
17	4913
18	5832

19	6859
20	8000
21	9261
22	10648
23	12167
24	13824
25	15625

Example

1,8,27,64,125,?

Next Cube

216

✓ Answer = **216**

Difference as Cubes

Example

1,2,10,37,101,?

Difference

1

8

27

64

Next difference

125

101+125

=226

✓ Answer = **226**

Alternative Series

Two different patterns run together.

Odd positions follow one rule.



Even positions follow another rule.

Example

5, 25, 7, 22, 9, 19, 11, ?

Odd Positions

5

7

9

11

(+2)

Even Positions

25

22

19

?

(-3)

19-3

=16

✓ Answer = 16



HOUSE OF
ASPIRANTS

Quick Exam Tricks

Trick 1

Always calculate the **difference first**. learn from someone who has already achieved what you are preparing for.

Trick 2

If differences are also changing,

Find the difference of differences (Second Difference).

Trick 3

If numbers increase rapidly,

Think about

- Multiplication
- Square
- Cube

Trick 4

If one operation doesn't work,

Try combinations

- $\times$ then $+$
- $\times$ then $-$
- $\div$ then $+$
- $\div$ then $-$

Trick 5

Check odd and even positions separately.

Many questions follow two different patterns.

Important Things to Memorize

Prime Numbers (1–100)

Squares (1–50)

Cubes (1–30)

Tables (2–50)

Final Exam Approach

1. **Find the Difference** (Difference Series?)
2. **Check Multiplication/Division** (Multiplication Series?)
3. **Look for Squares or Cubes**
4. **Check Prime Numbers**
5. **Check Alternate (Odd-Even) Pattern**



HOUSE OF
ASPIRANTS

PRACTICE SET

Question 1

What comes in place of (?)?

4, 8, 12, 16, 20, ?

- A) 22
- B) 24
- C) 26
- D) 28

Solution

Difference

+4
+4
+4
+4

Next

$$20 + 4 = 24$$

✓ Answer: B) 24

Question 2

What comes in place of (?)?

13, 18, 23, 28, 33, ?

- A) 36
- B) 37

C) 38

D) 39

Solution

Difference

+5
+5
+5
+5

$$33 + 5 = 38$$

✓ Answer: C) 38

Question 3

Find the missing number.

25, 31, 37, 43, 49, ?

- A) 53
- B) 54
- C) 55
- D) 56

Solution

Difference = +6

$$49 + 6 = 55$$

✓ Answer: C) 55

Question 4

Find the next term.

7, 14, 21, 28, 35, ?

- A) 40
- B) 41
- C) 42
- D) 43

Solution

Difference = +7

$$35 + 7 = 42$$

✓ Answer: C) 42

Question 5

Find the missing number.

56, 64, 72, 80, 88, ?

- A) 94
- B) 95
- C) 96
- D) 98

Solution

Difference = +8

$$88 + 8 = 96$$

✓ Answer: C) 96

★ Trick Box 1

If every difference is the same,

➡ It is a **Constant Addition Series**.

Question 6

Find the next number.

3, 5, 8, 12, 17, ?

- A) 22
- B) 23
- C) 24
- D) 25

Solution

Difference

+2

+3

+4

+5

Next

+6

$$17 + 6 = 23$$

✓ Answer: B) 23

Question 7

Find the missing number.

9, 11, 14, 18, 23, ?

- A) 28
- B) 29
- C) 30
- D) 31

Solution

Difference

+2

+3

+4

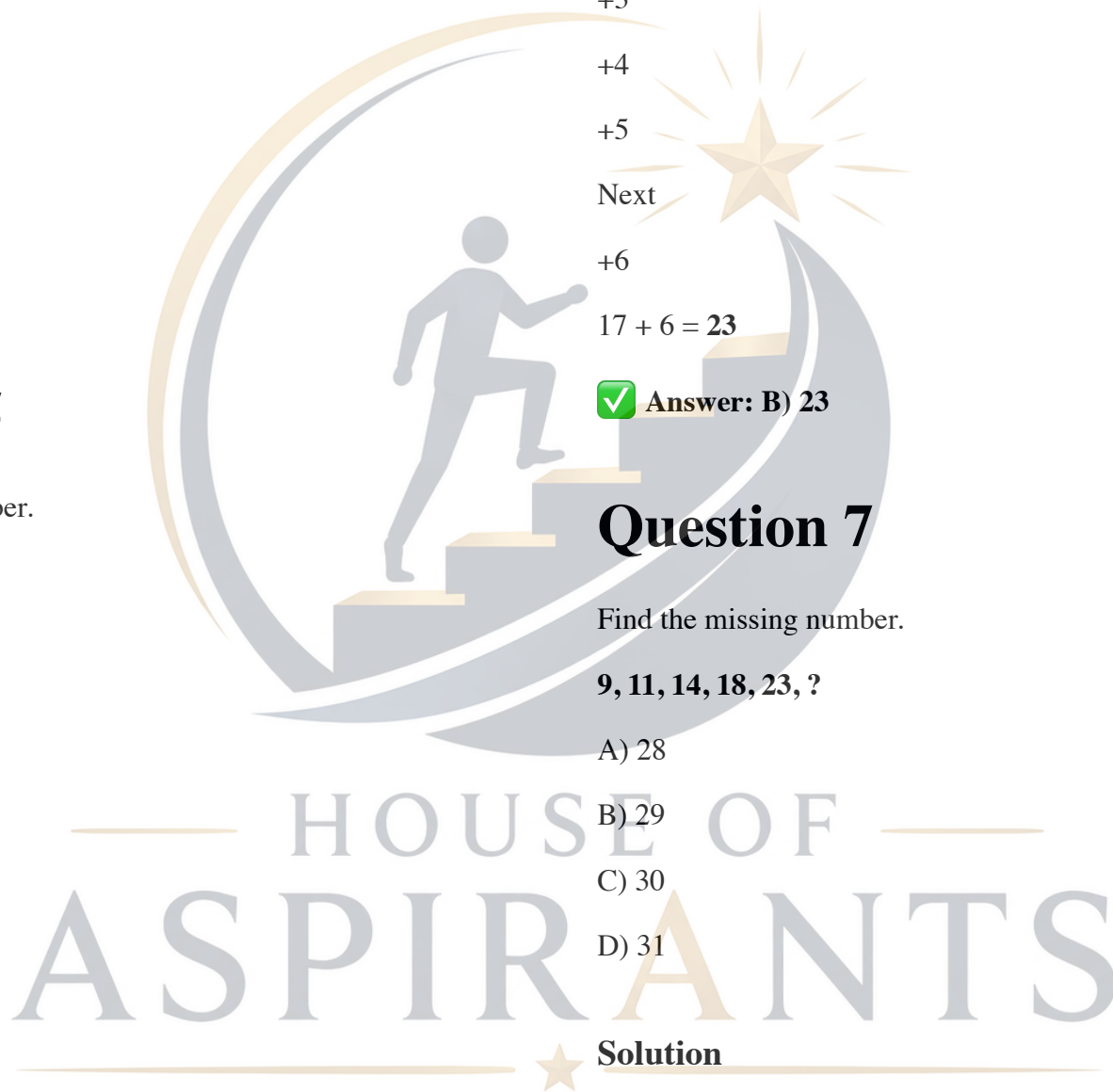
+5

Next = +6

$$23 + 6 = 29$$

✓ Answer: B) 29

Question 8



Find the next number.

15, 18, 22, 27, 33, ?

- A) 39
- B) 40
- C) 41
- D) 42

Solution

Difference

+3
+4
+5
+6

Next

+7

$$33 + 7 = 40$$

✓ Answer: B) 40

Question 9

Find the missing number.

30, 34, 39, 45, 52, ?

- A) 59
- B) 60
- C) 61
- D) 62

Solution

Difference

+4
+5
+6
+7

Next

+8

$$52 + 8 = 60$$

✓ Answer: B) 60

Question 10

Find the next term.

50, 55, 61, 68, 76, ?

- A) 83
- B) 84
- C) 85
- D) 86

Solution

Difference

+5
+6
+7
+8

Next

+9

$$76 + 9 = 85$$

✓ Answer: C) 85

★ Trick Box 2

If the difference increases by **1 every time**, then simply continue the pattern.

Question 11

Find the missing number.

60, 55, 50, 45, 40, ?

- A) 35
- B) 36
- C) 37
- D) 38

Solution

Difference

-5

-5

-5

-5

$$40 - 5 = 35$$

✓ Answer: A) 35

Question 12

Find the next term.

95, 90, 85, 80, 75, ?

A) 69

B) 70

C) 71

D) 72

Solution

Difference

-5

$$75 - 5 = 70$$

✓ Answer: B) 70

Question 13

Find the missing number.

84, 78, 72, 66, 60, ?

A) 55

B) 54

C) 53

D) 52

Solution

Difference

-6

$$60 - 6 = 54$$

✓ Answer: B) 54

Question 14

Find the next number.

100, 96, 92, 88, 84, ?

A) 79

B) 80

C) 81

D) 82

Solution

Difference

-4

$$84 - 4 = 80$$

✓ Answer: B) 80

Question 15

Find the missing number.

72, 65, 58, 51, 44, ?

A) 37

B) 38

C) 39

D) 40

Solution

Difference

-7

$$44 - 7 = 37$$

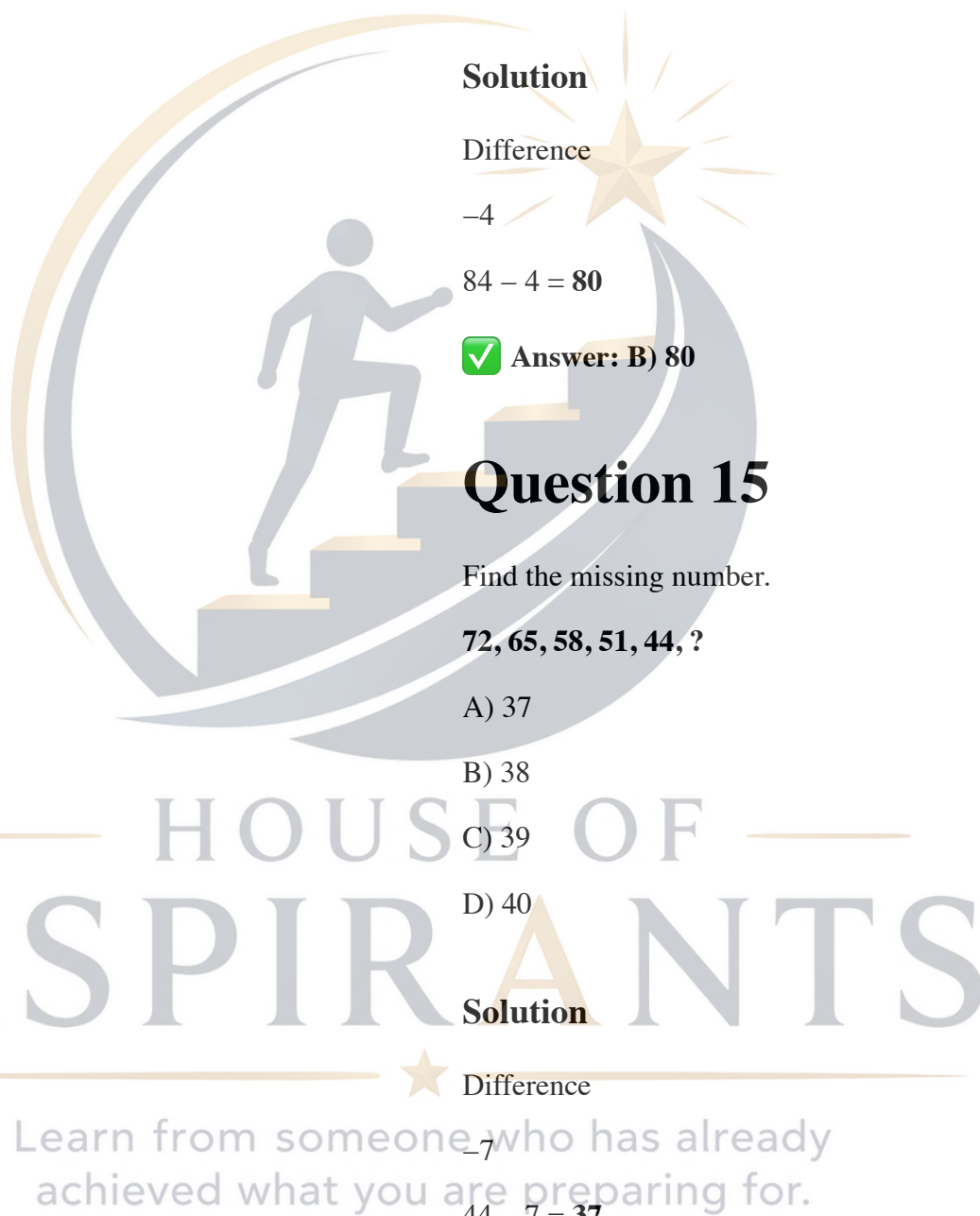
✓ Answer: A) 37

★ Trick Box 3

If every number decreases equally,

➡ It is a **Constant Subtraction Series**.

Question 16



Find the next term.

40, 38, 35, 31, 26, ?

- A) 19
- B) 20
- C) 21
- D) 22

Solution

Difference

- 2
- 3
- 4
- 5

Next

-6

$$26 - 6 = 20$$

☒ Answer: B) 20

Question 17

Find the missing number.

90, 89, 87, 84, 80, ?

- A) 73
- B) 74
- C) 75
- D) 76

Solution

Difference

- 1
- 2
- 3
- 4

Next

-5

$$80 - 5 = 75$$

☒ Answer: C) 75

Question 18

Find the next number.

75, 72, 68, 63, 57, ?

- A) 49
- B) 50
- C) 51
- D) 52

Solution

Difference

- 3
- 4
- 5
- 6

Next

-7

$$57 - 7 = 50$$

☒ Answer: B) 50

Question 19

Find the missing term.

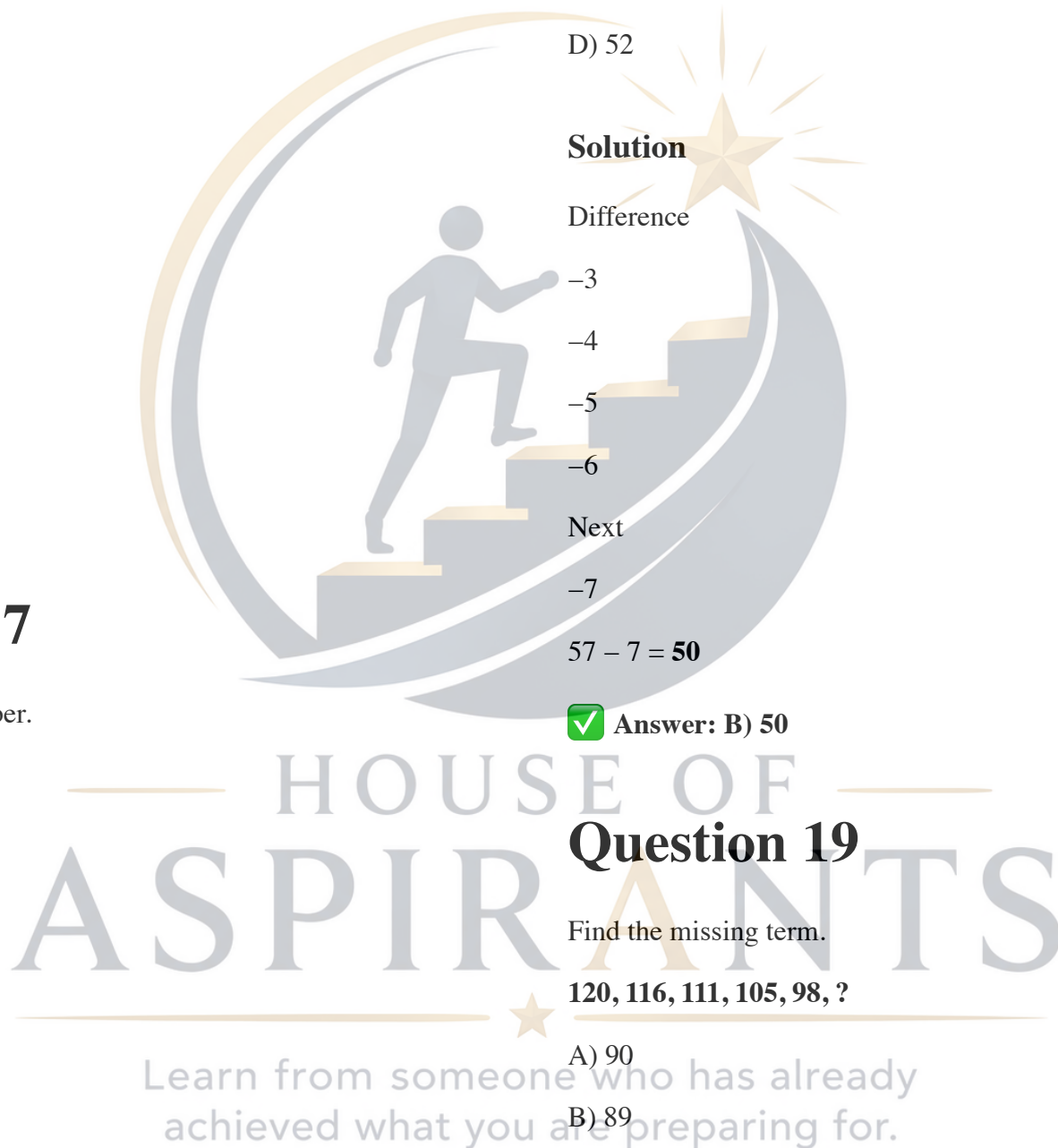
120, 116, 111, 105, 98, ?

- A) 90
- B) 89
- C) 88
- D) 87

Solution

Difference

- 4
- 5
- 6
- 7



Next

-8

$$98 - 8 = 90$$

✓ Answer: A) 90

Question 20

Find the next term.

150, 145, 139, 132, 124, ?

A) 115

B) 116

C) 117

D) 118

Solution

Difference

-5

-6

-7

-8

Next

-9

$$124 - 9 = 115$$

✓ Answer: A) 115

Prime Number Series

Question 21

What comes in place of (?)?

2, 3, 5, 7, 11, ?

A) 12

B) 13

C) 15

D) 17

Solution

These are consecutive prime numbers.

Next prime after 11 = **13**

✓ Answer: B) 13

Question 22

Find the missing number.

11, 13, 17, 19, 23, ?

A) 25

B) 27

C) 29

D) 31

Solution

Consecutive prime numbers.

Next = **29**

✓ Answer: C) 29

Question 23

Find the next number.

29, 31, 37, 41, 43, ?

A) 45

B) 47

C) 49

D) 51

Solution

Next prime after 43 = **47**

✓ Answer: B) 47

Question 24

Find the missing term.

53, 59, 61, 67, 71, ?

A) 73

B) 75

C) 77

D) 79

Solution

Next consecutive prime = **73**

✓ **Answer: A) 73**

Question 25

Find the next number.

79, 83, 89, 97, ?

A) 99

B) 101

C) 103

D) 107

Solution

Next prime after 97 = **101**

✓ **Answer: B) 101**

Trick Box 1

If every number is prime,

➡ Check the next consecutive prime number.

Prime Difference Series**Question 26**

Find the missing number.

10, 12, 15, 20, 27, ?

A) 36

B) 37

C) 38

D) 39

Solution

Difference

+2

+3

+5

+7

Next prime difference = +11

$27 + 11 = 38$

✓ **Answer: C) 38**

Question 27

Find the next number.

20, 23, 28, 35, 46, ?

A) 57

B) 59

C) 61

D) 63

Solution

Difference

+3

+5

+7

+11

Next prime = +13

$46 + 13 = 59$

✓ **Answer: B) 59**

Question 28

Find the missing number.

5, 7, 12, 19, 30, ?

A) 41

B) 43

C) 45

D) 47

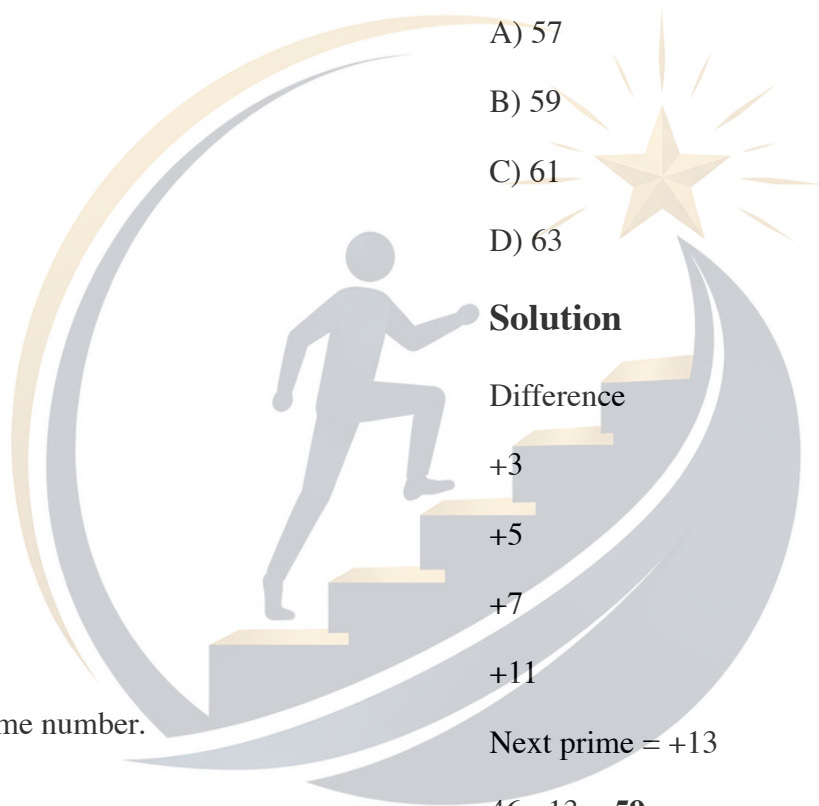
Solution

Difference

+2

+5

+7



HOUSE OF ASPIRANTS

Learn from someone who has already achieved what you are preparing for.

+11

Next prime = +13

$$30 + 13 = 43$$

✓ Answer: B) 43

Question 29

Find the next term.

100, 102, 105, 110, 117, ?

A) 126

B) 128

C) 130

D) 132

Solution

Difference

+2

+3

+5

+7

Next = +11

$$117 + 11 = 128$$

✓ Answer: B) 128

Question 30

Find the missing number.

50, 53, 58, 65, 76, ?

A) 87

B) 89

C) 91

D) 93

Solution

Difference

+3

+5

+7

+11

Next

+13

$$76 + 13 = 89$$

✓ Answer: B) 89

Trick Box 2

Sometimes the **numbers are not prime**,
but the **difference is prime**.

Always check both.

Square Series

Question 31

Find the next number.

1, 4, 9, 16, 25, ?

A) 36

B) 49

C) 64

D) 81

Solution

Squares

 1^2 2^2 3^2 4^2 5^2 Next = $6^2 = 36$

✓ Answer: A) 36

Question 32

Find the missing term.

16, 25, 36, 49, 64, ?

A) 72

- B) 81
C) 90
D) 100

Solution

Squares

4^2

5^2

6^2

7^2

8^2

Next = $9^2 = 81$

✓ Answer: B) 81**Question 33**

Find the next number.

49, 64, 81, 100, ?

- A) 111
B) 121
C) 144
D) 169

Solution

Squares

7^2

8^2

9^2

10^2

Next = 11^2

= 121

✓ Answer: B) 121**Question 34**

Find the missing number.

81, 100, 121, 144, 169, ?

- A) 196

- B) 200

- C) 210

- D) 225

Solution

Squares

9^2

10^2

11^2

12^2

13^2

Next

$14^2 = 196$

✓ Answer: A) 196**Question 35**

Find the next term.

225, 256, 289, 324, 361, ?

- A) 380
B) 400
C) 420
D) 441

Solution

Squares

15^2

16^2

17^2

18^2

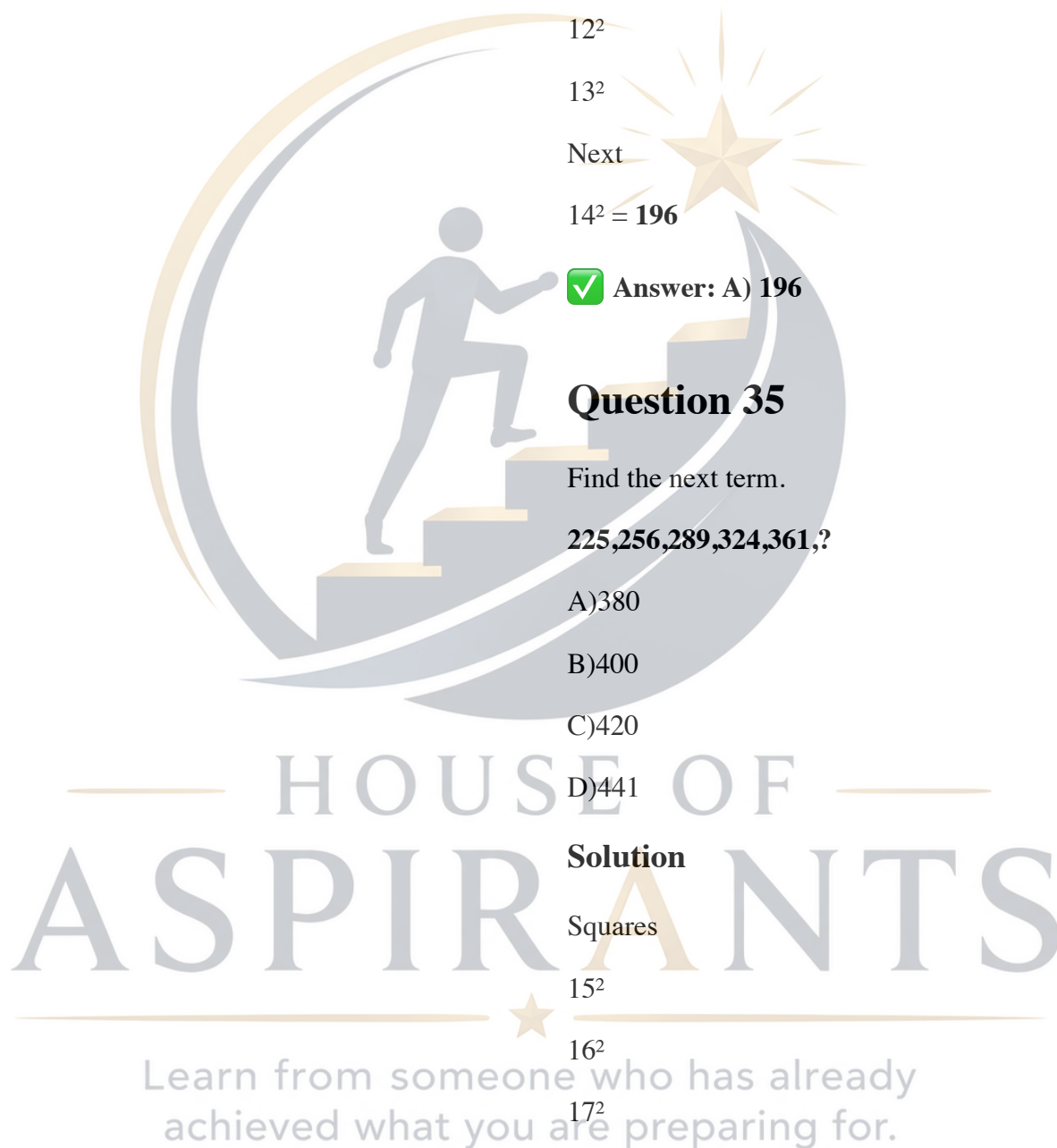
19^2

Next

$20^2 = 400$

✓ Answer: B) 400**Trick Box 3**

If numbers look familiar,



Think

$1^2, 2^2, 3^2, \dots$

or

$10^2, 11^2, 12^2, \dots$

Difference of Squares

Question 36

Find the next number.

2, 3, 7, 16, 32, ?

A) 57

B) 58

C) 59

D) 60

Solution

Difference

1

4

9

16

25

$32 + 25$

= 57

☒ Answer: A) 57

Question 37

Find the missing term.

5, 9, 18, 34, 59, ?

A) 95

B) 94

C) 93

D) 92

Solution

Difference

4

9

16

25

Next

36

$59 + 36$

= 95

☒ Answer: A) 95

Question 38

Find the next number.

10, 11, 15, 24, 40, ?

A) 60

B) 64

C) 65

D) 66

Solution

Difference

1

4

9

16

Next

25

$40 + 25$

= 65

☒ Answer: C) 65

Cube Series

Question 39

Find the next number.

1, 8, 27, 64, 125, ?

- A)196
B)216
C)225
D)243

Solution

Consecutive cubes

1^3

2^3

3^3

4^3

5^3

Next

6^3

$= 216$

✓ Answer: B) 216**Question 40**

Find the missing term.

1,2,10,37,101,?

- A)216
B)221
C)226
D)231

Solution

Difference

1

8

27

64

Next difference

125

$101+125$

$= 226$

✓ Answer: C) 226

Multiplication Series

Question 41

What comes in place of (?)?

4, 12, 36, 108, 324, ?

- A) 648
B) 729
C) 864
D) 972

SolutionEach number is multiplied by **3**.

$4 \times 3 = 12$

$12 \times 3 = 36$

$36 \times 3 = 108$

$108 \times 3 = 324$

$324 \times 3 = 972$

✓ Answer: D) 972**Question 42**

Find the missing number.

5, 20, 80, 320, ?

- A) 960
B) 1280
C) 1440
D) 1600

SolutionMultiply by **4**

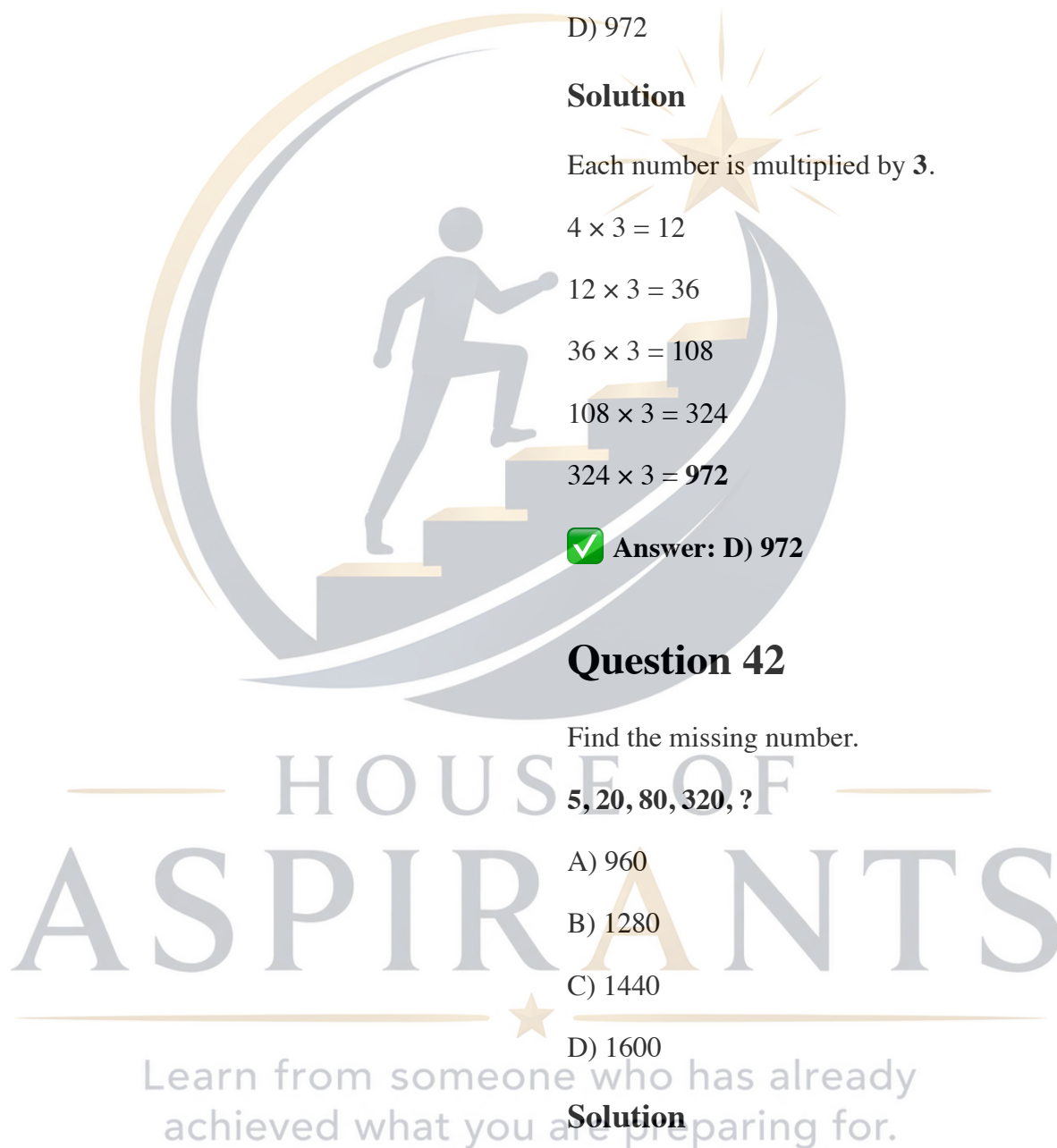
$320 \times 4 = 1280$

✓ Answer: B) 1280**Question 43**

Find the next number.

6, 18, 54, 162, ?

- A) 324



B) 486

C) 648

D) 972

Solution

Multiply by 3

$$162 \times 3 = 486$$

✓ Answer: B) 486

Question 44

Find the missing number.

7, 35, 175, 875, ?

A) 3500

B) 4375

C) 5250

D) 6125

Solution

Multiply by 5

$$875 \times 5 = 4375$$

✓ Answer: B) 4375

Question 45

Find the next term.

9, 27, 81, 243, ?

A) 486

B) 729

C) 972

D) 999

Solution

Multiply by 3

$$243 \times 3 = 729$$

✓ Answer: B) 729

Trick Box 1

If numbers increase very rapidly,

Always check multiplication first.

Multiplication + Addition**Question 46**

Find the missing number.

5, 11, 23, 47, 95, ?

A) 190

B) 191

C) 192

D) 193

Solution

Rule

$$\times 2 + 1$$

$$95 \times 2 + 1$$

$$= 191$$

✓ Answer: B) 191

Question 47

Find the next number.

8, 17, 35, 71, 143, ?

A) 286

B) 287

C) 288

D) 289

Solution

Rule

$$\times 2 + 1$$

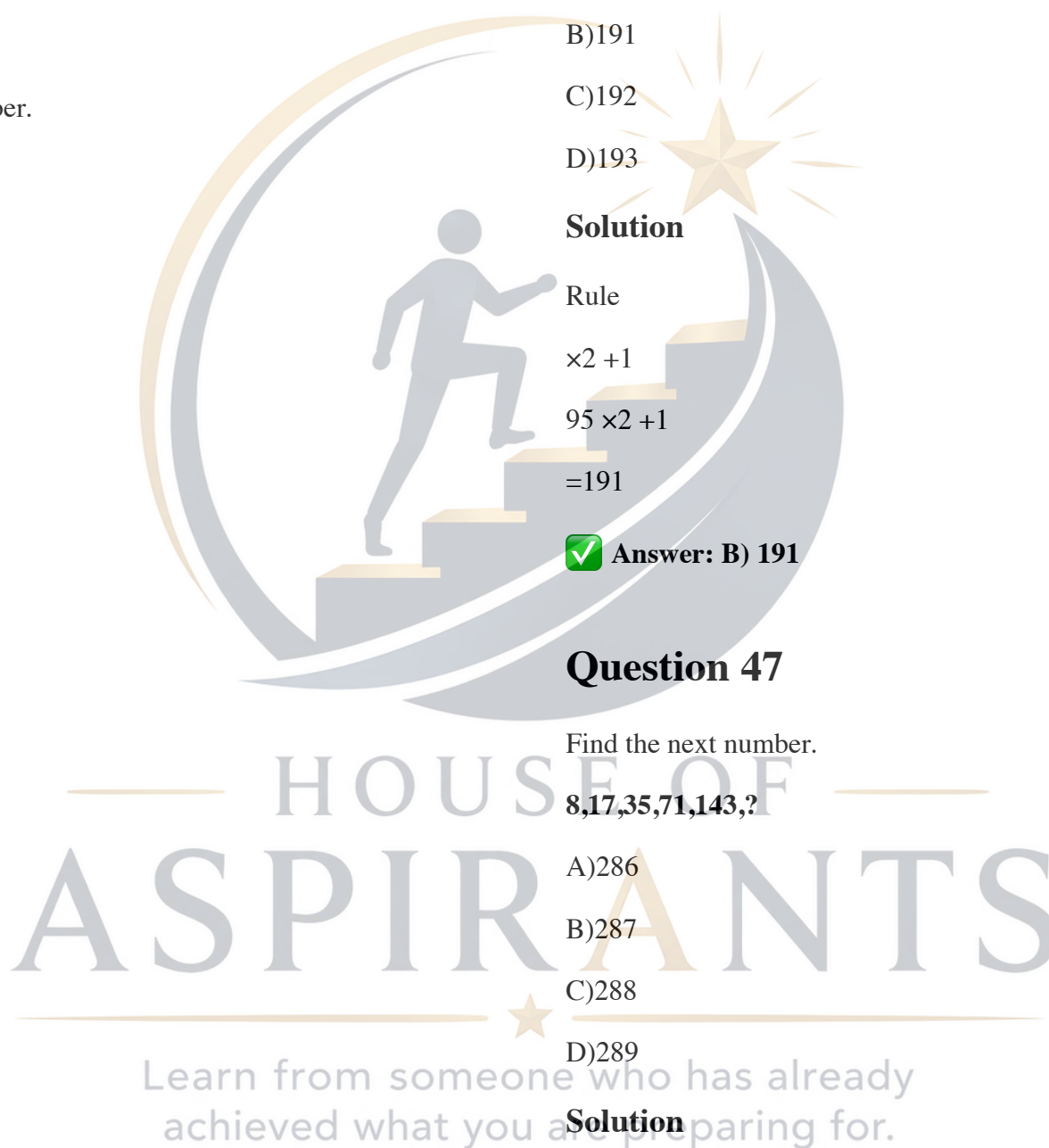
$$143 \times 2 + 1$$

$$= 287$$

✓ Answer: B) 287

Question 48

Find the missing term.



10,21,43,87,175,?

A)349

B)350

C)351

D)352

Solution

Rule

$\times 2 + 1$

$175 \times 2 + 1$

$= 351$

✓ Answer: C) 351

Question 49

Find the next number.

3,7,15,31,63,?

A)126

B)127

C)128

D)129

Solution

Rule

$\times 2 + 1$

$63 \times 2 + 1$

$= 127$

✓ Answer: B) 127

Question 50

Find the missing term.

6,13,27,55,111,?

A)222

B)223

C)224

D)225

Solution

Rule

$\times 2 + 1$

$111 \times 2 + 1$

$= 223$

✓ Answer: B) 223

Trick Box 2

The most common SSC/Punjab Police pattern is

Next Number = Previous $\times 2 + 1$

Always test this first.

Multiplication + Consecutive Addition

Question 51

5,11,24,51,106,?

A)211

B)212

C)213

D)217

Solution

Rule

$\times 2 + 1$

$\times 2 + 2$

$\times 2 + 3$

$\times 2 + 4$

Next

$106 \times 2 + 5$

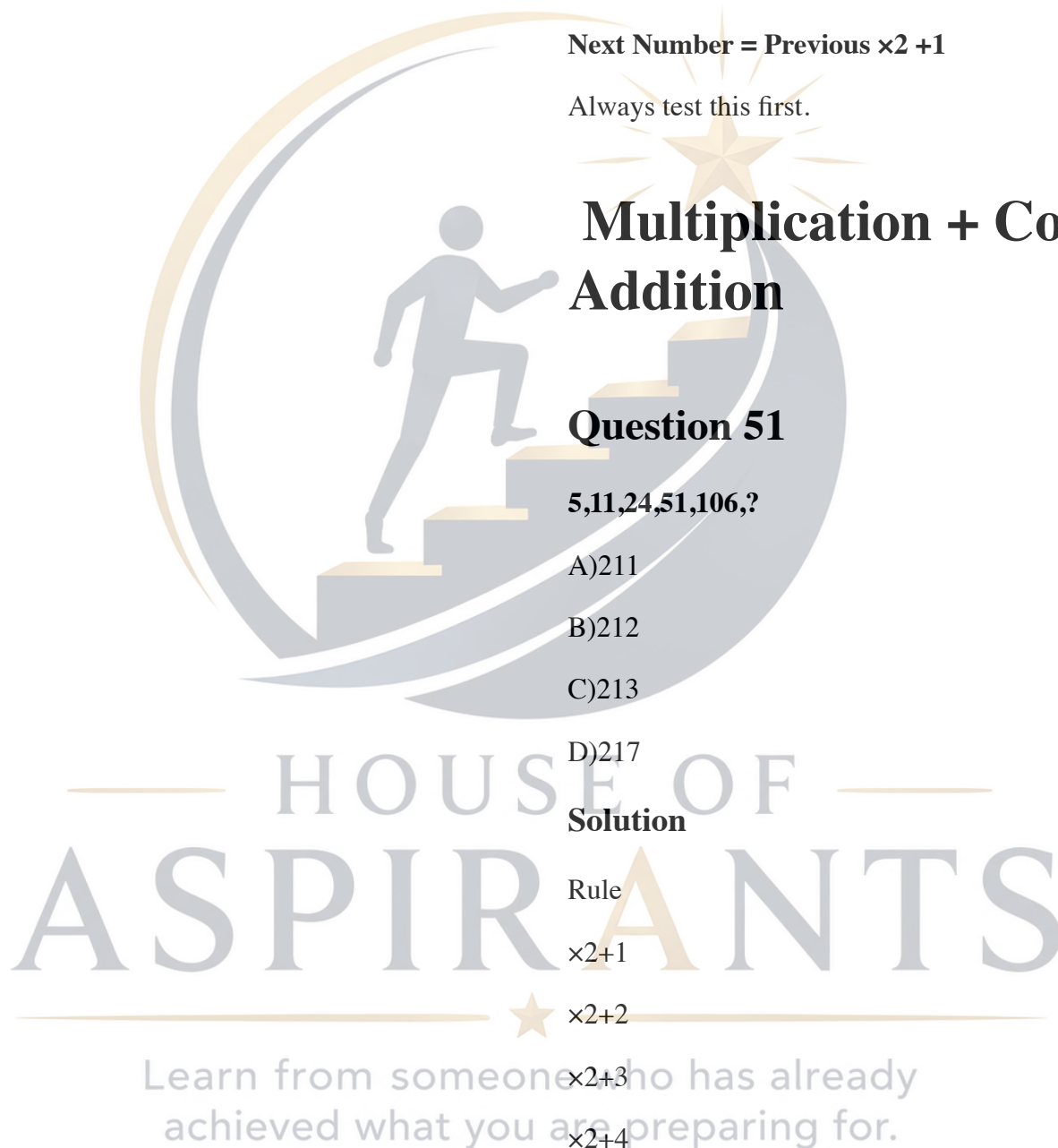
$= 217$

✓ Answer: D) 217

Question 52

4,9,20,43,90,?

A)182



B)183

C)185

D)186

Solution

Rule

 $\times 2 + 1$ $\times 2 + 2$ $\times 2 + 3$ $\times 2 + 4$

Next

 $90 \times 2 + 5$ $= 185$  **Answer: C) 185****Question 53****6,13,28,59,122,?**

A)245

B)246

C)249

D)250

Solution

Rule

 $\times 2 + 1$ $\times 2 + 2$ $\times 2 + 3$ $\times 2 + 4$

Next

 $122 \times 2 + 5$ $= 249$  **Answer: C) 249****Question 54****7,15,32,67,138,?**

A)279

B)280

C)281

D)282

Solution $138 \times 2 + 5$ $= 281$  **Answer: C) 281****Question 55****9,19,40,83,170,?**

A)340

B)342

C)345

D)347

Solution $170 \times 2 + 7$

Wait!

Check carefully:

 $9 \times 2 + 1 = 19$ $19 \times 2 + 2 = 40$ $40 \times 2 + 3 = 83$ $83 \times 2 + 4 = 170$ $170 \times 2 + 5 = 345$  **Answer: C) 345****Trick Box 3**

When the answer is close,

Check whether

 $+1, +2, +3, +4 \dots$

has been added after multiplication.

 Division Series**Question 56**

336,168,84,42,21,?

- A)10
- B)10.5
- C)11
- D)12

Solution

Divide by 2

$$21 \div 2$$

$$=10.5$$

✓ Answer: B) 10.5

Question 57

7200,3600,1200,300,60,?

- A)10
- B)12
- C)15
- D)20

Solution

$$\div 2$$

$$\div 3$$

$$\div 4$$

$$\div 5$$

$$\div 6$$

$$60 \div 6$$

$$=10$$

✓ Answer: A) 10

Question 58

640,320,160,80,40,?

- A)10
- B)20
- C)25
- D)30

Solution

Divide by 2

$$40 \div 2$$

$$=20$$

✓ Answer: B) 20

Mixed Multiplication & Division

Question 59

24,72,36,108,54,?

- A)150
- B)156
- C)160
- D)162

Solution

$$\times 3$$

$$\div 2$$

$$\times 3$$

$$\div 2$$

$$\times 3$$

$$54 \times 3$$

$$=162$$

✓ Answer: D) 162

Question 60

16,48,24,72,36,?

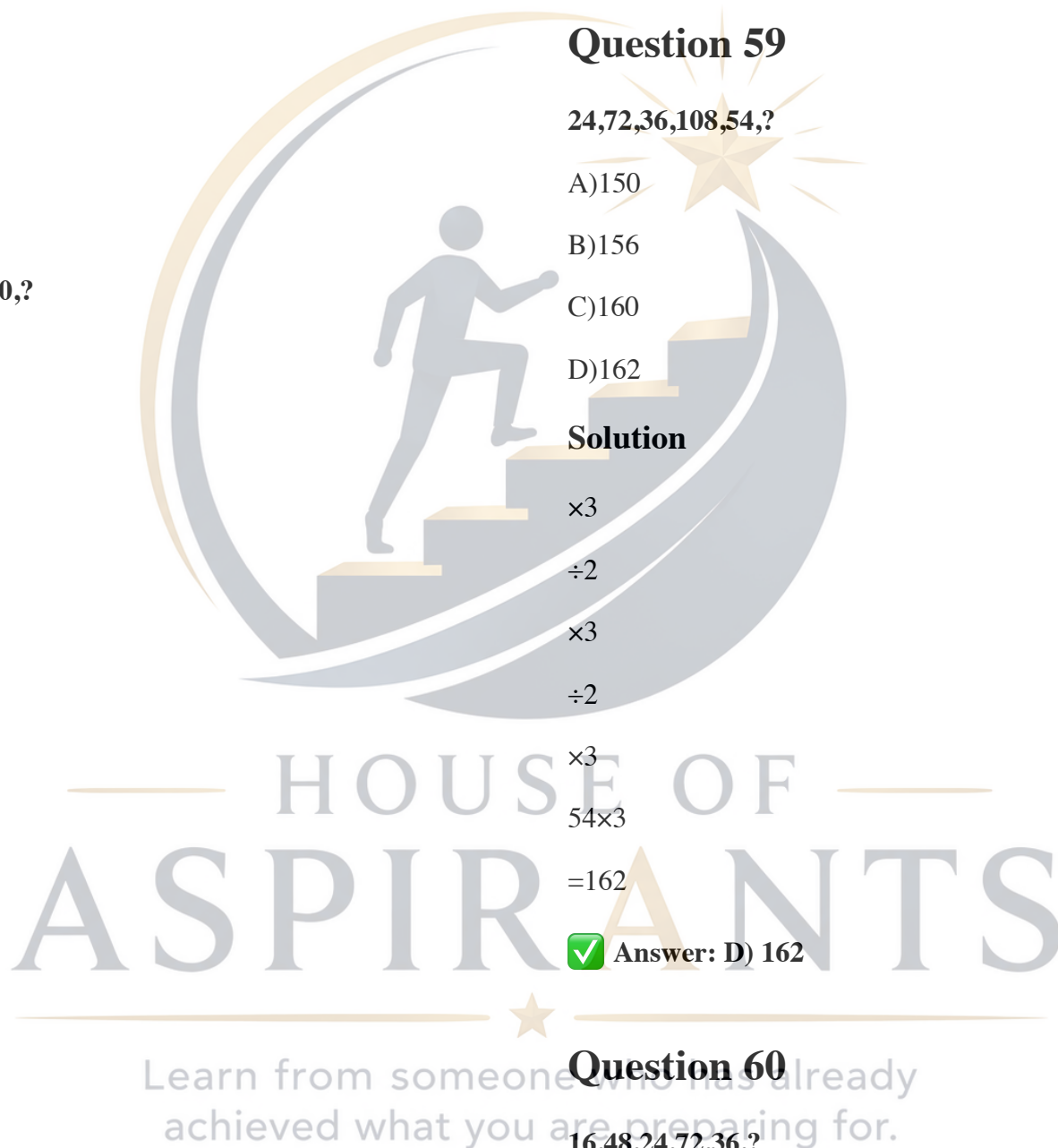
- A)98
- B)104
- C)108
- D)112

Solution

$$\times 3$$

$$\div 2$$

$$\times 3$$



$$\div 2$$

$$\times 3$$

$$36 \times 3$$

$$= 108$$

✓ Answer: C) 108

Alternative Series (Odd & Even Positions)

Question 61

What comes in place of (?)?

5, 25, 7, 22, 9, 19, 11, ?

- A) 14
- B) 15
- C) 16
- D) 17

Solution

Separate the series:

Odd Positions:
5, 7, 9, 11 (+2)

Even Positions:
25, 22, 19, ? (-3)

$$19 - 3 = 16$$

✓ Answer: C) 16

Question 62

Find the missing number.

4, 30, 6, 27, 8, 24, 10, ?

- A) 18
- B) 20
- C) 21
- D) 22

Solution

Odd Positions:
4, 6, 8, 10 (+2)

Even Positions:
30, 27, 24, ? (-3)

$$24 - 3 = 21$$

✓ Answer: C) 21

Question 63

Find the next term.

2, 40, 5, 36, 8, 32, 11, ?

- A) 26
- B) 27
- C) 28
- D) 29

Solution

Odd Positions:
2, 5, 8, 11 (+3)

Even Positions:
40, 36, 32, ? (-4)

$$32 - 4 = 28$$

✓ Answer: C) 28

Question 64

Find the missing number.

10, 50, 15, 45, 20, 40, 25, ?

- A) 30
- B) 35
- C) 36
- D) 38

Solution

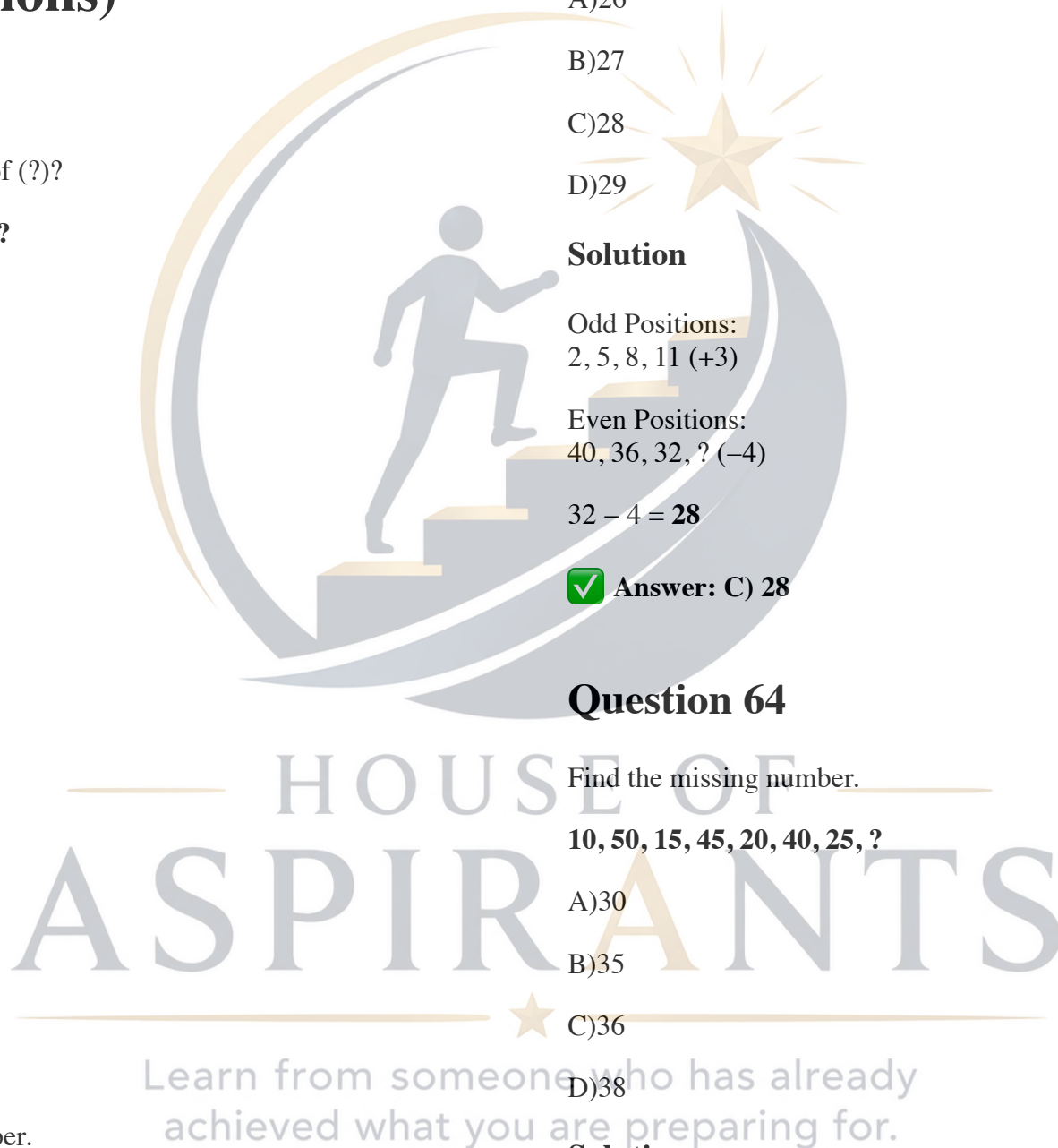
Odd:
10, 15, 20, 25 (+5)

Even:
50, 45, 40, ? (-5)

$$40 - 5 = 35$$

✓ Answer: B) 35

Question 65



Find the next number.

3, 21, 6, 18, 9, 15, 12, ?

A)10

B)11

C)12

D)13

Solution

Odd:

3, 6, 9, 12 (+3)

Even:

21, 18, 15, ? (-3)

15 - 3 = 12

✓ Answer: C) 12

Trick Box 1

Whenever a series looks confusing,

👉 Separate **Odd Position Numbers** and **Even Position Numbers**.

Most alternative series become easy after this.

Fibonacci Pattern

(Next Number = Sum of Previous Two Numbers)

Question 66

1, 1, 2, 3, 5, 8, 13, ?

A)18

B)20

C)21

D)22

Solution

13 = 5 + 8

Next = 8 + 13 = 21

✓ Answer: C) 21

Question 67

2, 3, 5, 8, 13, 21, ?

A)30

B)32

C)34

D)34

Solution

13 + 21

= 34

✓ Answer: C) 34

Question 68

4, 5, 9, 14, 23, 37, ?

A)55

B)58

C)60

D)62

Solution

23 + 37

= 60

✓ Answer: C) 60

Mixed Difference Series

Question 69

3, 7, 13, 21, 31, ?

A)42

B)43

C)44

D)45

Solution

Difference

4

6

8

10

Next =12

 $31+12=43$ ☒ Answer: B) 43**Question 70**

5,10,18,29,43,?

A)58

B)60

C)61

D)62

Solution

Difference

5

8

11

14

Next=17

 $43+17$ $=60$ ☒ Answer: B) 60**Trick Box 2**

If the first difference isn't constant,

Find the pattern in the differences.

Square + Addition Pattern**Question 71**

2,5,10,17,26,?

A)35

B)36

C)37

D)38

Solution

Difference

3

5

7

9

Next=11

 $26+11$ $=37$

OR

Series = n^2+1 ☒ Answer: C) 37**Question 72**

3,6,11,18,27,?

A)36

B)37

C)38

D)39

Solution

Difference

3

5

7

9

Next

11

 $27+11$ $=38$ ☒ Answer: C) 38**Question 73**

6,9,14,21,30,?

A)39

B)40

C)41

D)42

Solution

Difference

3

5

7

9

Next

11

30+11

=41**✓ Answer: C) 41****Cube Difference Pattern****Question 74****1,9,36,100,225,?**

A)441

B)441

C)441

D)441

Solution

Difference

8

27

64

125

(Cubes)

Next Difference

216

225+216

=441**✓ Answer: A) 441****Question 75****2,10,37,101,226,?**

A)442

B)442

C)443

D)444

Solution

Difference

8

27

64

125

Next

216

226+216

=442**✓ Answer: A) 442****Trick Box 3**

Difference

1,8,27,64,125...

Means

 $1^3, 2^3, 3^3, 4^3, 5^3$ Learn from someone who has already
achieved what you are preparing for.**Advanced Mixed Pattern****Question 76****6,12,24,48,96,?**

A)180

B)182

C)190

D)192

Solution

$$\times 2$$

$$96 \times 2$$

$$= 192$$

✓ Answer: D) 192

Question 77

7, 15, 31, 63, 127, ?

A) 253

B) 254

C) 255

D) 256

Solution

$$\times 2 + 1$$

$$127 \times 2 + 1$$

$$= 255$$

✓ Answer: C) 255

Question 78

100, 95, 85, 70, 50, ?

A) 25

B) 20

C) 18

D) 15

Solution

Difference

$$-5$$

$$-10$$

$$-15$$

$$-20$$

Next

$$-25$$

$$50 - 25$$

$$= 25$$

✓ Answer: A) 25

Question 79

2, 6, 18, 54, 162, ?

A) 324

B) 405

C) 486

D) 540

Solution

$$\times 3$$

$$162 \times 3$$

$$= 486$$

✓ Answer: C) 486

Question 80

3, 9, 27, 81, 243, ?

A) 486

B) 648

C) 729

D) 972

Solution

$$\times 3$$

$$243 \times 3$$

$$= 729$$

✓ Answer: C) 729

Question 81

Find the missing number.

7, 15, 31, 63, 127, ?

A) 254

B) 255

C) 256

D) 257



Solution

Pattern:

$\times 2 + 1$

$127 \times 2 + 1$

$= 255$

✓ Answer: B) 255**Question 82**

Find the next term.

3, 6, 12, 24, 48, ?

A) 72

B) 84

C) 96

D) 98

SolutionEach number $\times 2$

$48 \times 2 = 96$

✓ Answer: C) 96**Question 83**

Find the missing number.

5, 10, 18, 29, 43, ?

A) 58

B) 60

C) 62

D) 64

Solution

Difference

$+5$

$+8$

$+11$

$+14$

Next = $+17$

$43 + 17$

$= 60$

✓ Answer: B) 60**Question 84**

Find the next term.

1, 4, 9, 16, 25, ?

A) 36

B) 42

C) 49

D) 64

Solution

Perfect Squares

1^2

2^2

3^2

4^2

5^2

Next

6^2

$= 36$

✓ Answer: A) 36**Question 85**

Find the missing number.

2, 3, 5, 8, 13, 21, ?

A) 30

B) 32

C) 34

D) 35

Solution

Fibonacci Pattern

$13 + 21$

$= 34$

✓ Answer: C) 34

Trick Box 1

Whenever the series doesn't follow addition or multiplication,

Check whether

Next = Previous Two Numbers Added

Question 86

Find the next number.

64, 32, 16, 8, 4, ?

A) 1

B) 2

C) 3

D) 4

Solution

$\div 2$

$4 \div 2$

= 2

✓ Answer: B) 2

Question 87

Find the missing number.

2, 6, 18, 54, 162, ?

A) 324

B) 405

C) 486

D) 540

Solution

$\times 3$

162×3

= 486

✓ Answer: C) 486

Question 88

Find the next term.

100, 95, 85, 70, 50, ?

A) 20

B) 25

C) 30

D) 35

Solution

Difference

-5

-10

-15

-20

Next

-25

$50 - 25$

= 25

✓ Answer: B) 25

Question 89

Find the missing number.

1, 8, 27, 64, 125, ?

A) 196

B) 216

C) 225

D) 243

Solution

Perfect Cubes

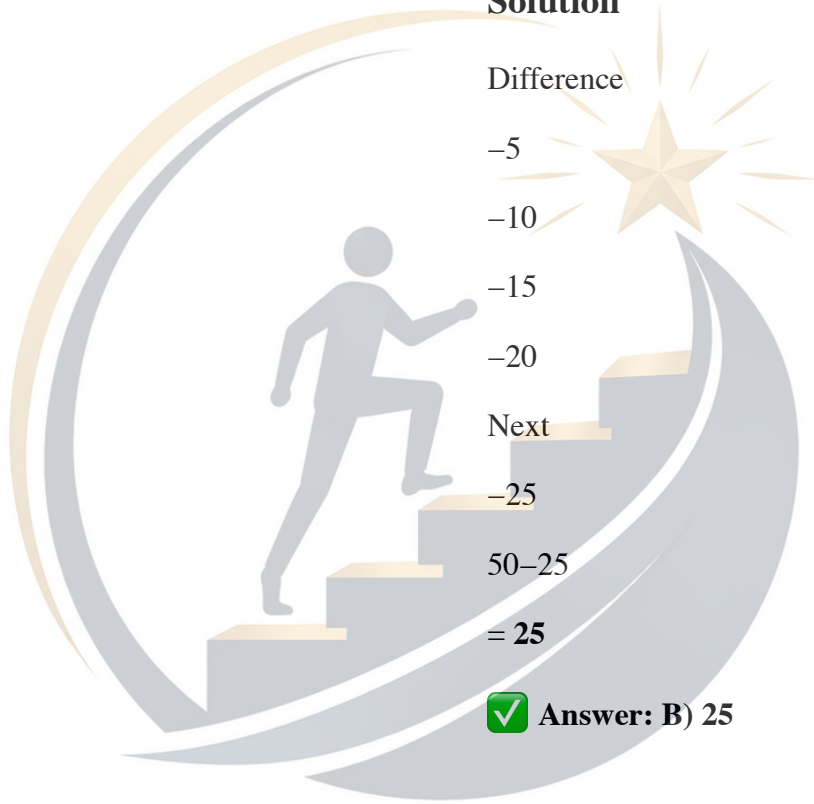
1^3

2^3

3^3

4^3

5^3



HOUSE OF
ASPIRANTS

Learn from someone who has already
achieved what you are preparing for.

Next

$$6^3$$

$$= 216$$

✓ Answer: B) 216

Question 90

Find the next term.

4, 9, 16, 25, 36, ?

A) 45

B) 47

C) 49

D) 64

Solution

Squares

$$2^2$$

$$3^2$$

$$4^2$$

$$5^2$$

$$6^2$$

Next

$$7^2$$

$$= 49$$

✓ Answer: C) 49

Trick Box 2

If numbers look familiar,

Always compare them with

- Squares
- Cubes
- Prime Numbers

Question 91

Find the next number.

24, 72, 36, 108, 54, ?

A) 150

B) 156

C) 160

D) 162

Solution

$$\times 3$$

$$\div 2$$

$$\times 3$$

$$\div 2$$

$$\times 3$$

$$54 \times 3$$

$$= 162$$

✓ Answer: D) 162

Question 92

Find the missing number.

5, 25, 7, 22, 9, 19, 11, ?

A) 14

B) 15

C) 16

D) 17

Solution

Odd

$$5, 7, 9, 11 (+2)$$

Even

$$25, 22, 19, ?$$

$$+3$$

$$19 - 3$$

$$= 16$$

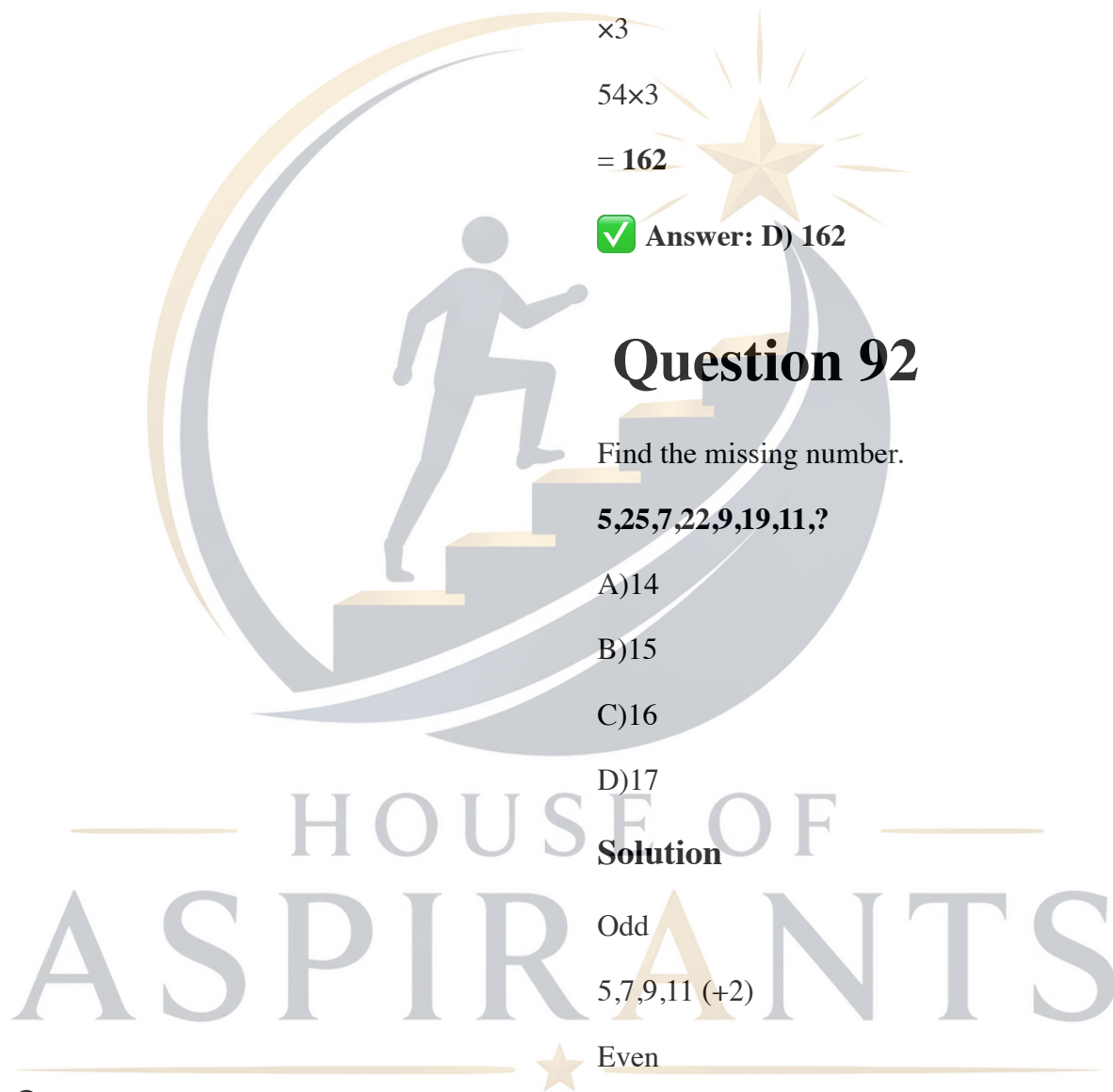
✓ Answer: C) 16

Question 93

Find the next term.

11, 13, 17, 19, 23, ?

A) 25



B)27

C)29

D)31

Solution

Prime Numbers

Next = 29

✓ Answer: C) 29**Question 94**

Find the missing number.

2,3,7,16,32,?

A)56

B)57

C)58

D)59

Solution

Difference

1

4

9

16

25

32+25

= 57**✓ Answer: B) 57****Question 95**

Find the next term.

3,7,15,31,63,?

A)126

B)127

C)128

D)129

Solution $\times 2 + 1$ $63 \times 2 + 1$ **= 127****✓ Answer: B) 127****Trick Box 3**

The most repeated SSC/Punjab Police pattern is

Previous $\times 2 + 1$ **Question 96**

Find the missing number.

336,168,84,42,21,?

A)10

B)10.5

C)11

D)12

Solution $\div 2$ $21 \div 2$ **= 10.5****✓ Answer: B) 10.5****Question 97**

Find the next number.

10,12,15,20,27,?

A)36

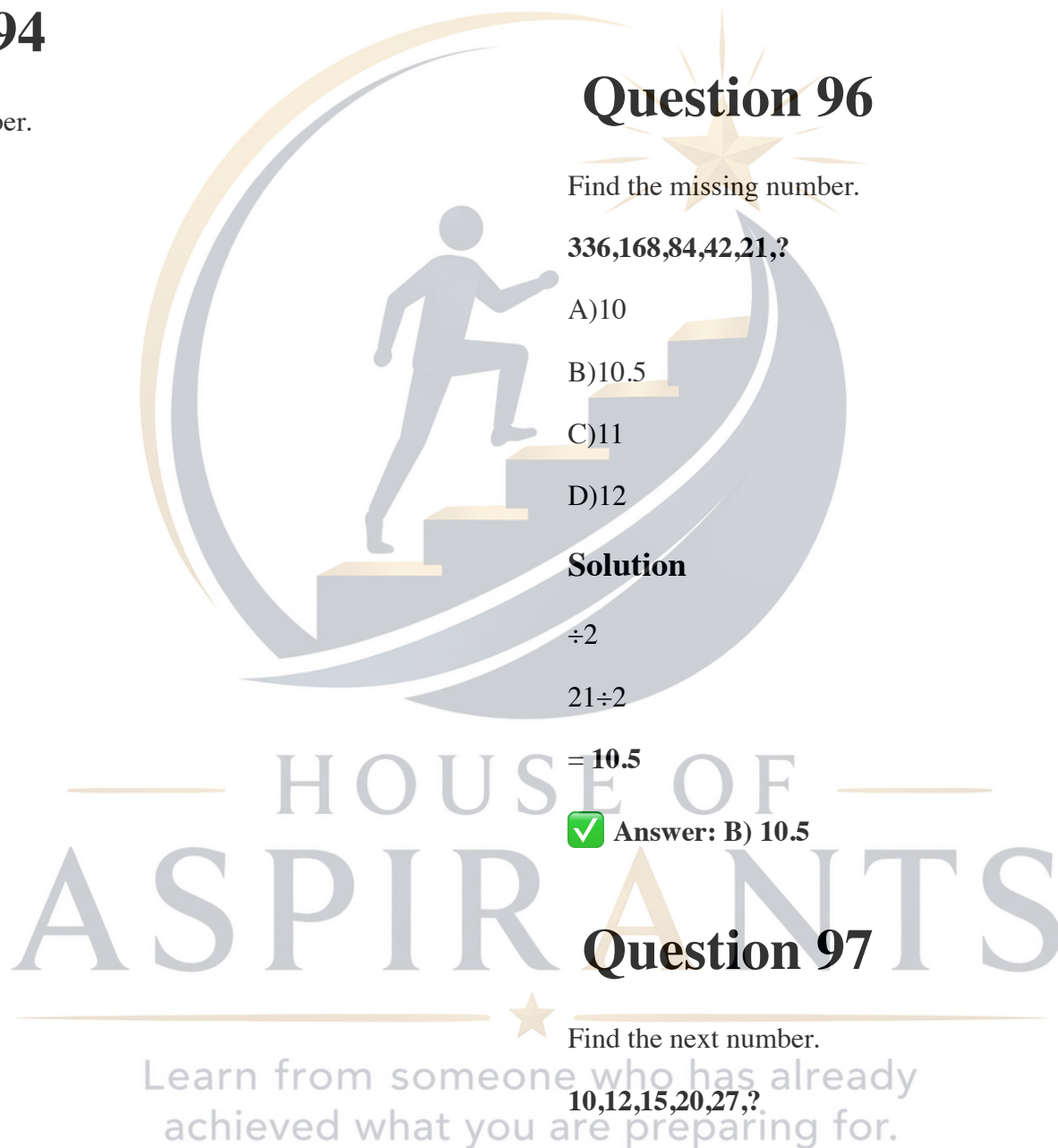
B)37

C)38

D)39

Solution

Difference

+2**+3**

+5
+7
Next
+11
27+11
= **38**

✓ Answer: C) 38

Question 98

Find the missing number.

150,145,139,132,124,?

- A)114
- B)115
- C)116
- D)117

Solution

Difference

-5
-6
-7
-8
Next
-9
124-9
= **115**

✓ Answer: B) 115

Question 99

Find the next term.

5,11,24,51,106,?

- A)211
- B)213
- C)215
- D)217

Solution

$\times 2 + 1$
 $\times 2 + 2$
 $\times 2 + 3$
 $\times 2 + 4$
Next
 $106 \times 2 + 5$
= **217**

✓ Answer: D) 217

Question 100

Find the missing number.

1,2,10,37,101,?

- A)216
- B)221
- C)226
- D)231

Solution

Difference

1
8
27
64
125

(Cubes)

$101 + 125$

= **226**

✓ Answer: C) 226

